

Grade 8 ICT – Number Systems (Notes)

1. Number Systems

A **number system** is a method used to represent numbers using a fixed set of symbols.

Computers store and process all types of data (text, images, audio, video, and numbers) in **binary form (0s and 1s)**.

2. Types of Number Systems

There are four common number systems.

Number System	Symbols	Base
Binary	0, 1	2
Octal	0, 1, 2, 3, 4, 5, 6, 7	8
Decimal	0, 1, 2, 3, 4, 5, 6, 7, 8, 9	10
Hexadecimal (Hex)	0–9, A, B, C, D, E, F	16

Note: In **Grade 8**, only the **Binary** and **Decimal** number systems are studied.

3. Base (Radix) of a Number System

The **base (radix)** indicates the number of different symbols used in a number system.

Examples:

- Binary $\rightarrow 101_2$
- Decimal $\rightarrow 231_{10}$
- Octal $\rightarrow 573_8$
- Hexadecimal $\rightarrow 7B_{16}$

The base is written as a **subscript**.

4. Decimal Number System (Base 10)

The decimal number system is the number system used in everyday life.

Symbols

0, 1, 2, 3, 4, 5, 6, 7, 8, 9

Base

10

Place Values in Decimal Numbers

Each digit has a place value.

Place	Power	Value
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Hundreds	10^2	100
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Tens	10^1	10
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Ones	10^0	1
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Example

Convert **231₁₀** into expanded form.

231₁₀

$$= (2 \times 10^2) + (3 \times 10^1) + (1 \times 10^0)$$

$$= (2 \times 100) + (3 \times 10) + (1 \times 1)$$

$$= 200 + 30 + 1$$

$$= \mathbf{231}$$

5. Binary Number System (Base 2)

The binary number system uses only **two symbols**.

Symbols

0 and 1

Base

2

Each binary digit is called a **Bit**.

Bit = Binary Digit

Place Values in Binary Numbers

Place Power Value

2^0	1
2^1	2
2^2	4
2^3	8
2^4	16
2^5	32
2^6	64
2^7	128
2^8	256
2^9	512
2^{10}	1024

Example

Convert **101₂** into decimal.

101₂

$$= (1 \times 2^2) + (0 \times 2^1) + (1 \times 2^0)$$

$$= (1 \times 4) + (0 \times 2) + (1 \times 1)$$

$$= 4 + 0 + 1$$

$$= \mathbf{5_{10}}$$

6. Why Number System Conversion is Important

Computers understand only **binary numbers (0 and 1)**.

When a user enters data:

- Numbers
- Letters

- Images
- Sounds
- Videos

the computer converts them into **binary**, processes them, and converts the result back into a form humans can understand.

Example:

User types **9**

↓

Computer converts it to **00001001**

↓

CPU processes the binary data

↓

Monitor displays **9**

7. Converting Decimal to Binary

Steps

1. Divide the decimal number by **2**.
2. Write down the **remainder**.
3. Continue dividing the quotient by **2** until the quotient becomes **0**.
4. Read the remainders **from bottom to top**.

Example 1

Convert **13₁₀** to binary.

Division Quotient Remainder

13 ÷ 2	6	1
6 ÷ 2	3	0
3 ÷ 2	1	1

Division Quotient Remainder

$$1 \div 2 \quad 0 \quad 1$$

Read from bottom to top:

1101_2

Example 2

Convert **34_{10}** to binary.

Division Quotient Remainder

$$34 \div 2 \quad 17 \quad 0$$

$$17 \div 2 \quad 8 \quad 1$$

$$8 \div 2 \quad 4 \quad 0$$

$$4 \div 2 \quad 2 \quad 0$$

$$2 \div 2 \quad 1 \quad 0$$

$$1 \div 2 \quad 0 \quad 1$$

Answer:

100010_2

Example 3

Convert **49_{10}** to binary.

Division Quotient Remainder

$$49 \div 2 \quad 24 \quad 1$$

$$24 \div 2 \quad 12 \quad 0$$

$$12 \div 2 \quad 6 \quad 0$$

$$6 \div 2 \quad 3 \quad 0$$

$$3 \div 2 \quad 1 \quad 1$$

Division Quotient Remainder

$$1 \div 2 \quad 0 \quad 1$$

Answer:

110001₂

8. Converting Binary to Decimal

Steps

1. Write the place values (powers of 2).
 2. Multiply each binary digit by its place value.
 3. Add all the values.
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Example 1

Convert **10111₂** to decimal.

$$10111_2$$

$$= (1 \times 16) + (0 \times 8) + (1 \times 4) + (1 \times 2) + (1 \times 1)$$

$$= 16 + 0 + 4 + 2 + 1$$

$$= \mathbf{23_{10}}$$

Example 2

Convert **10010₂** to decimal.

$$10010_2$$

$$= (1 \times 16) + (0 \times 8) + (0 \times 4) + (1 \times 2) + (0 \times 1)$$

$$= 16 + 2$$

$$= \mathbf{18_{10}}$$

Example 3

Convert **11111₂** to decimal.

$$11111_2$$

$$= 16 + 8 + 4 + 2 + 1$$

$$= 31_{10}$$

Example 4

Convert **10101000₂** to decimal.

$$10101000_2$$

$$= (1 \times 128) + (0 \times 64) + (1 \times 32) + (0 \times 16) + (1 \times 8) + (0 \times 4) + (0 \times 2) + (0 \times 1)$$

$$= 128 + 32 + 8$$

$$= 168_{10}$$

9. Binary Representation of Data

A computer stores all information using **ON** and **OFF** electronic signals.

State Binary

OFF 0

ON 1

Everything inside the computer is represented using binary, including:

- Numbers
 - Letters
 - Text
 - Images
 - Audio
 - Videos
-

10. Data Processing in a Computer

The flow of data inside a computer is:

Input Device

↓

Convert data into Binary



Main Memory



CPU Processes Binary Data



Main Memory



Output Device

11. Comparison of Binary and Decimal Number Systems

Feature	Binary	Decimal
Base	2	10
Symbols	0, 1	0–9
Example	110101 ₂	217 ₁₀
Used By	Computers	Humans